Prove that: $\log_{\sqrt{2}}(9)$ is irrational

<u>Proof (by contradiction)</u>

Assume, for contradiction, that $\log_{\sqrt{2}} 9$ is rational

then $\log_{\sqrt{2}} 9 = \frac{a}{b}$ where $a, b \in \mathbb{Z}$, $b > 0$, and $\gcd(a,b) = 1$

$$\log_{\sqrt{2}} 9 = \frac{a}{b}$$

$$\Leftrightarrow (\sqrt{2})^{a/b} = 9$$

$$\Leftrightarrow (\sqrt{2})^{a} = 9^{b} \quad \text{or} \quad 2^{a/2} = 9^{b}$$

Square both sides: $2^{a} = 9^{2b}$

$2^{a} = 3^{4b}$ Impossible to find a and b

because 2^{a} has only the prime of 2
3^{4b} has only the prime of 3

By the Fundamental Theorem of Arithmetic, a positive integer has a unique prime factorization.

The assumption, "$\log_{\sqrt{2}} 9 = \frac{a}{b}$" leads to $2^{a} = 3^{4b}$ impossible

The contration shows our assumption was false.

$\log_{\sqrt{2}} 9$ is irrational.